11 B Internal energy is the sum of a random distribution of kinetic and potential energies associated with the molecules of a system.

A is false during the change in states of the matter. However, the converse is true; i.e. when the temperature increases, internal energy increases.

B is true when work is done on the system.

C is false in general.
Consider the two separate gases (e.g. hydrogen and helium) having different amount in moles but same internal energy, they have different average kinetic energies, and hence different temperatures.

D is false. It should be the change (increase) of internal energy, not the internal energy itself.

12 C $\frac{1}{2} m c^2 = \frac{3}{2} kT$, where T is thermodynamic temperature, c is the r.m.s. speed